XK-JS1-1

Robotic Arm Training Device



Instruction Book

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Chapter 1 System Introduction

1.1 Product Overview:

This manipulator with three degrees of freedom, that's X、Y.、Z, the X axis for front and back movement, the Y axis for up and down movement, Z axis for rotation movement; There is also another claw for grasping action. X、Y、Z are all adopt stepper motor drive, the transmission mechanism of X axis and Y axis used lead screw drive、Z axis used gear drive; The claw is drived by electromagnet. This device can realize grab、level lift and transport action etc.

This training device is developed according to the teaching requirement of mechanical and electrical integration. It is one set of teaching training device that integrated the PLC control、positioning control. PLC chooses Omron Siemens S7-200 series programmable controller, the motors of X axis and Y axis used 57 stepper motor, Z axis chooses 42 stepper motor.

1.2 Technical Parameter

A. Input power: single phase three wires～220V±10% 50Hz

B. Working environment: temperature -10℃～+40℃, relative humidity <85％ (25℃), height<4000m.

C. Device capacity: <250VA

D. Earth leakage protection action current: ≤30mA; Earth leakage protection action time: ≤0.1s

E. Size: 900×640×1200mm

F. X axle journey: 200mm

G. Y axle journey: 440mm

H. Y axle rotation angle: 360°

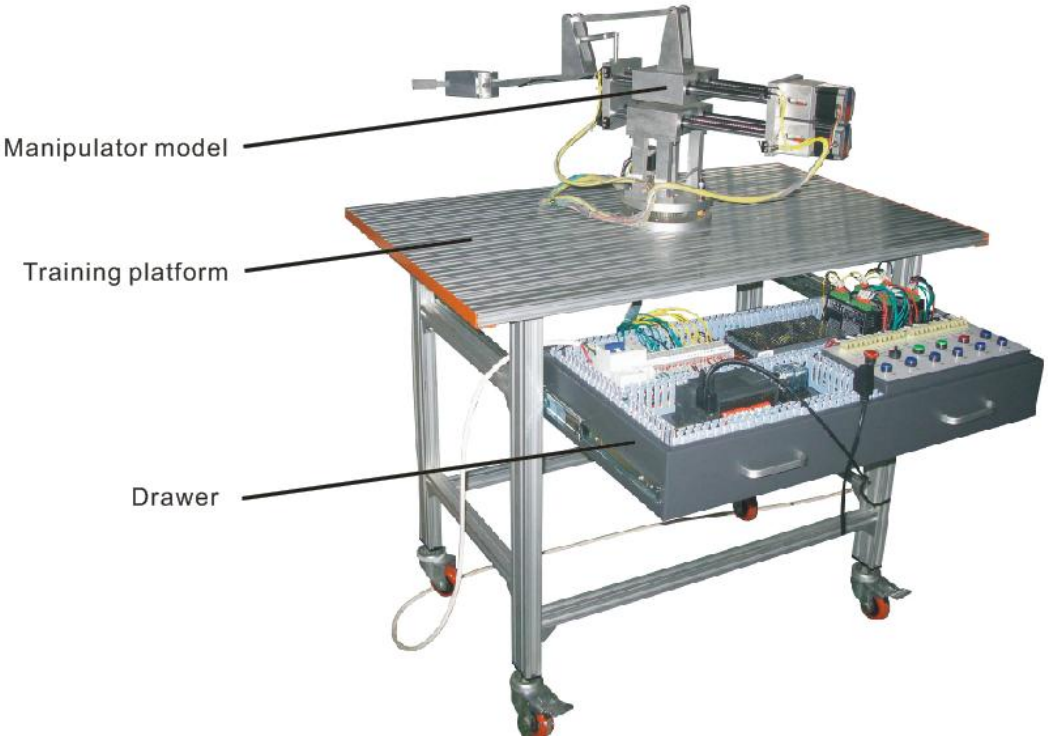
9、Y 轴旋转角度:360°

1.3 Product Structure

1.3.1 Device overview picture

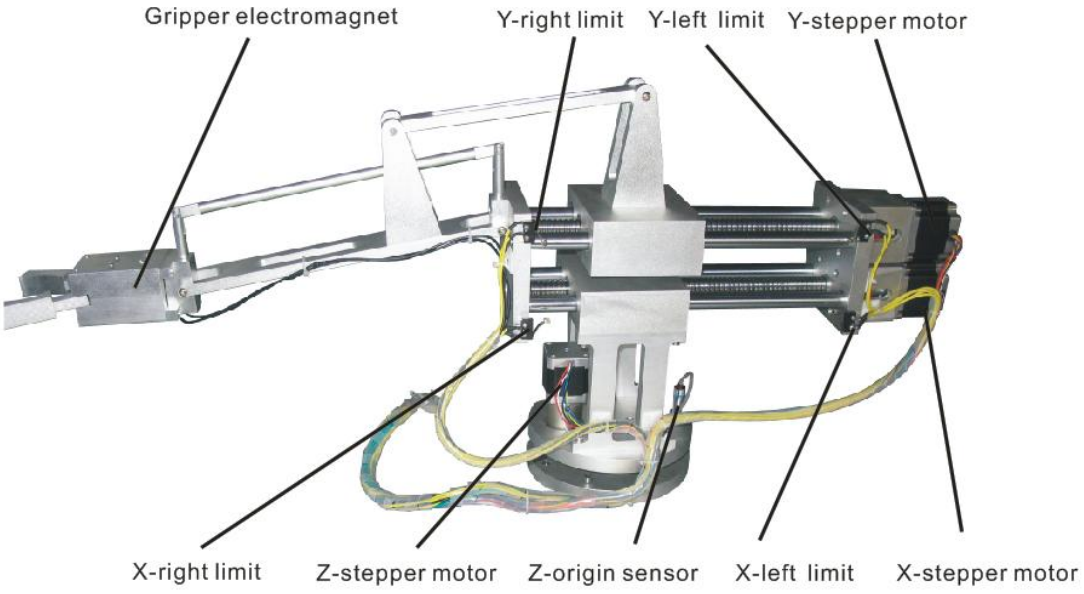


1.3.2 Whole mechanism structure

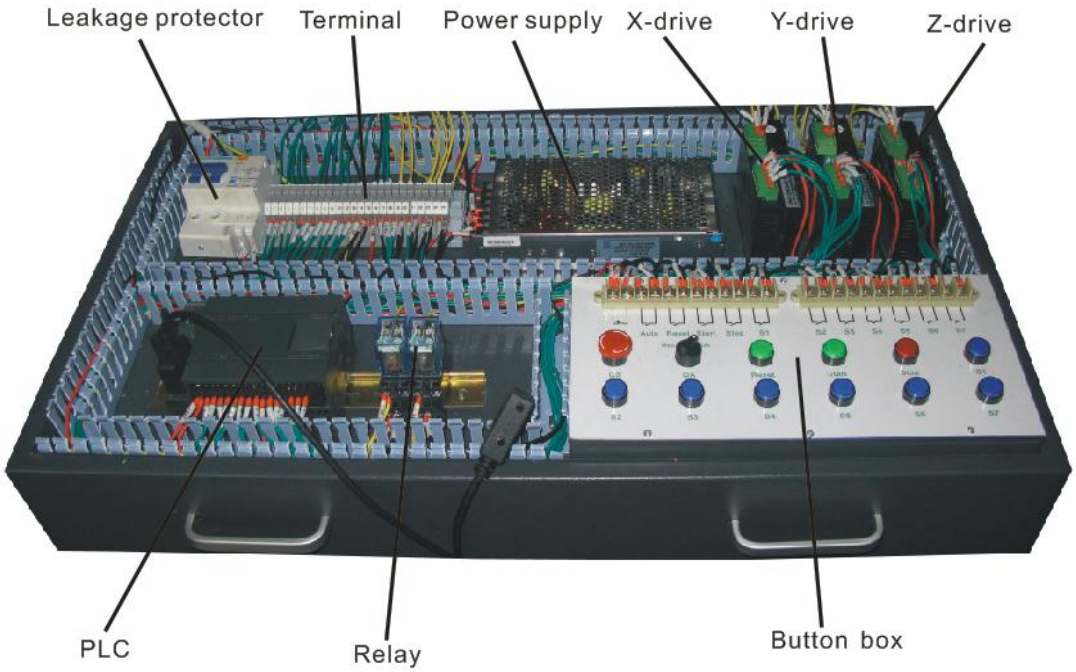


1.3.3 Unit structure

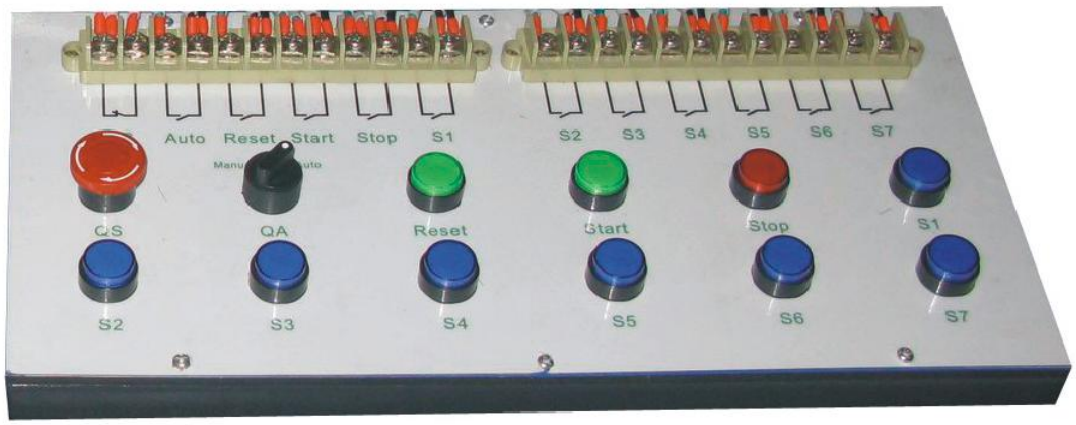
A. Manipulator model structure



B. Drawer structure



C. Button box structure



QS: Emergency button, used for PLC power control;

QA: Automatic、manual selector switch, when switch to left side is for manual, to right side is automatically;

Reset: Reset button;

Start: Auto start button;

Stop: Auto stop button;

S1: The claw firmly grasp、loose inching button;

S2: Manipulator left rotation button;

S3: Manipulator right rotation button;

S4: X axle forward movement button;

S5: X axle backward movement button;

S6: Y axle upward movement button;

S7: Y axle downward movement button.

Chapter 2 Electrical Control System

2.1 Stepper motor and actuator

Stepper motor is the kind of special operating motor that converts the electrical pulses signal to the corresponding angular displacement or linear displacement. For input one electrical pulse signal one time, motor will rotate one angle. Its movement type is stepper, so called the stepper motor

2.1.1 Working principle of stepper motor

The following takes one most simple three phase reaction type stepper motor as an example, introduce the working principle of stepper motor.

Figure 2-1 is principle diagram of one set three phase reaction type stepper motor.

Stator core is salient pole type, a total of three pairs (quantity of 6) magnetic pole, every two spaces on the relative polarity be around a phase control winding. The rotor made of soft magnetic materials, is also salient pole structure, there are only four teeth, tooth width is equal to the polar wide of stator.

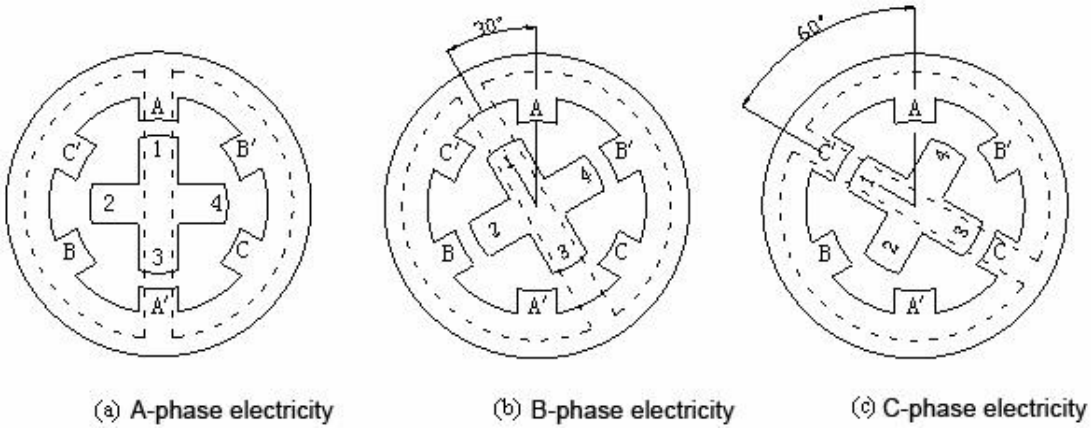


Figure 2-1

When A phase control winding get power and the rest two phase are not get electricity, in the motor, set up the magnetic field that takes the A phase polar as the axis.

Because the magnetic flux with the characteristics that pass the magnetic resistance minimum path, make the axis of the rotor teeth 1、3, and stator A phase polar axis alignment, show in Figure 2-1 (a). If A phase control winding outage、B phase control winding get power, rotor under the action of reaction torque, counterclockwise turn s around 30 °, make the axis of rotator teeth 2、4 and stator B phase polar axis alignment, that's mean the rotor go one step, show in Figure 2-1 (b). if disconnect B phase, make C phase control winding get electricity, rotor anti-clockwise turn 30°, make the axis of rotor teeth 1、3 and stator C phase polar axis alignment, show in Figure 2-1 (c).

So according to the order of A—B—C—A get power take turns, rotor will be counter-clockwise direction rotation step by step. Its speed based on the frequency that each phase control winding power on or off, rotation direction based on the order that control winding power on in turn. If power on according to order of A—C—B—A, so motor will be rotation with clockwise.

The above type of get power is three phase single three clap. "three phase" mean three phase stepper motor; "single three clap" means each time only one phase to control the winding get electricity; the control winding change power on or off state one time means one clap, "three clap" means change the power on or off state three times as one cycle. The angle that each clap rotor turning called the stepping angle. When the three phase

single three clap running, stepping angle is 30°. So, this angle is not big, cannot be real used.

If change the power on type of control winding to A→AB→B→BC→C→CA→A, namely that one phase get power then two phase get power and carry out interval in turns, finish one cycle require to change 6 times of power up state, called the three phase single、double six clap power up mode. When A、B two phase winding power on at same time, the rotor teeth position should be consider the two couple of stator pole action at same time, only the middle position that magnetic pull is balanced which produced by A phase and B phase for rotor teeth, this middle position will be the balanced position. So, power up state that under single、double six claps, the rotor balanced position increase twice, stepping angle is 15°

Further to reduce the stepping angle is to use the stator pole with a small tooth, the rotor teeth with many structures, analysis show that this structure of the stepper motor, the interval Angle can be very small. Generally speaking, the actual stepping motor products are always use this method to realize the subdivide of the stepping angle.

2.1.2 2M542-N stepper driver

A. Performance index

2M542-N is one kind of equal angle constant torque subdivided driver, driver voltage DC 24-58V, adapter current below 4.2A, outer diameter 42,57MM, the various types of two-phase hybrid stepping motor. This driver internal adopt the circuit that similar servo control principle, this circuit can make motor running steady with low speed, almost do not have vibration and noisy, motor running is steady, heating smaller, the stepper pulse stop more than 100m/s, driver current auto to reduce half. The positioning precision can up to 25600 steps/turning

B. Main Feature

24-50V DC power supply, suitable for the worst of electric net environment

H double polar constant phase current subdivide drive

Original speed self-adaptive circuit, automatic optimization

Subdivide current setting is convenient

2-64 subdivide 15 running mode

Maximum step no. 25600Pulse/rev

Over-current protection, over-voltage protection, under-voltage protection, short circuit protection

Off-line（ENS）protection function

Work peak current range: 1.0A-4.2A, output current total 8 grades.

Current resolution ratio is 0.45A

C. Electrical index

Explanation	Min value	Typical value	Max value
Supply voltage (VDC)	24	36V	50
Normal working output current (A) peak value	1.0	-----	4.2
Logic input current (mA)	7	10	16
Stopper pulse relative level rate (KHZ)	----	----	200
Pulse low level time (US)	2.5	----	----

D. Cooling type

Cooling type	Natural cooling or forced cooling	
Using environment	Situation	Avoid dust, oil mist, and corrosive gas
	Stored temperature	-10·C~80·C
	Max. environment temperature	65·C
	Environment temperature	＜ 80%RH, non condensation without frosting
Vibration	5.9m/s²Max	
Weight	0.26kg	

E. Current setting

Dial switch			Current value （A） Average value
SW1	SW2	SW3	
1	1	1	1.00
0	1	1	1.46
1	0	1	1.91
0	0	1	2.37
1	1	0	2.84
0	1	0	3.31
1	0	0	3.76
0	0	0	4.20

F. Subdivide setting

Dial switch				Subdivide sheet
SW4	SW5	SW6	SW7	
0	1	1	1	400
1	0	1	1	800
0	0	1	1	1600
1	1	0	1	3200
0	1	0	1	6400
1	0	0	1	12800
0	0	0	1	25600
1	1	1	0	1000
0	1	1	0	2000
1	0	1	0	4000
0	0	1	0	5000
1	1	0	0	8000
0	1	0	0	10000
1	0	0	0	20000
0	0	0	0	25000

G. Input signal interface description

Mark symbol	Function	Details
PLS+	Input signal optoelectronic isolation plus end	Connect +5V power supply, +5V~+24V all can drive, if higher than +5V required connect current-limiting resistance
PLS-	Stepper pulse signal	Falling edge effective, each time pulse from high to low, the motor goes further one step. Input resistance 220Ω , requirement: low level 0-0.5V, high level 4-5V, pulse width >2.5uS
DIR+	Input signal optoelectronic isolation plus end	Connect +5V supply power, +5V~+24V all can Be driven, higher than +5V required connect current –limiting resistance
DIR-	Direction control signal	Used for change direction of motor rotation, input resistance 220Ω , requirement: low level 0-0.5V, high level 4-5V, pulse width >2.5uS
ENA+	Input signal optoelectronic isolation plus end	Connect +5V supply power, +5V-+24V all can drive, higher than +5V require to connect current-limiting resistance
ENA-	Motor release signal	Effective (low level) turn off motor coil current, drive stop working, motor under the free state.

H. Motor wiring

The use of stepper motor, one is to pay attention the correct installation, the second is the correct wiring.

Installation of the stepper motor, It must be in strict accordance with the requirements of

the product description. Stepper motor is one kind of precision device, pay attention that, it's not allow to beat its shaft end during installation and even don't remove the motor.

Wiring is different for different stepper motor, for XK-JS1-1, the X axes and Y axes use the 57 stepper motor which is 8 wires system、Z axes 42 stepper motor is 6 wires system.

The wiring diagram of 6 wires system show in Figure 2-2 (a),

There are two kinds of wiring for 8 wires system: parallel connection and series connection, parallel wiring is high speed constant power connection method, show in Figure 2-2 (b), series wiring is high torque low speed connection method, show in Figure 2-2 （c）.

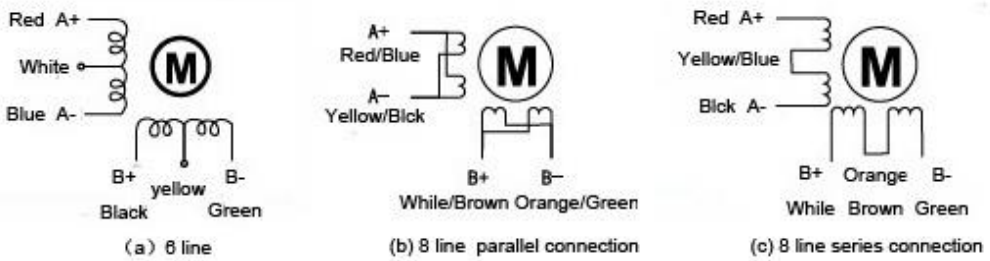


Figure 2-2

2.1.3 Note for stepper motor using

During control the running of stepper motor, it should pay attention that consider about prevent the step out of stepping motor running.

Stepper motor step out include lose step and over step. During lose step, the number of rotor forward steps is smaller than the pulse number, during the over step, the number of rotor forward steps is more than pulse number. When lose step is serious, will make the rotor stay in one place or vibration around one position; when the over step is serious, the device will be happen overshoot.

The operations that make manipulator return to the origin, usually happen the situation that over step. When the manipulator device back to the origin, the origin switch action, make the command input OFF. But the speed is too high before reaching the origin. The inertia torque will be bigger than the holding torque of stepping motor and make the stepper motor over step.

So the operation that go back to the origin should ensure enough low-speed advisable; When the stepper motor to drive the manipulator running at a high speed, it emergency stop, it's unavoidable to happen over step, therefore it should be first return to the origin with low speed, then restore the original method of the operation.

Due to the motor winding itself is the inductive load, the higher the input frequency, the less the exciting current. High frequency, magnetic flux change greatly, eddy current losses become large. Therefore, higher of input frequency, output torque is reduced.

Output torque of maximum working frequency can just achieve 40-50% of low frequency torque. During control of the high-speed positioning, if the specified frequency is too high, it will be happen the lose step phenomenon.

In addition, if the mechanical parts cannot adjust well, it can make the mechanical load increase. Stepper motor can't be load running, even a moment, it will happen out of step, When serious situation it will be halt or vibration origin irregularly over and over again.

2.2 S7-200 PLC Pulse output function and position control programming

S7-200 has two built-in PTO/PWM generator, to build high-speed pulse sequence (PTO) or pulse width adjustment (PWM) signal waveform. A generator is assigned to the digital output point Q0.0, another generator assigned to digital output point Q0.1

When the configuration one output is PTO operation, to generate a 50% duty cycle pulse sequence is used for stepping motor or servo motor speed and location open loop control. Built-in PTO provides pulse string output, pulse cycle and quantity can be controlled by the user. But the application must be through the PLC built-in I/O to provide direction and limit control.

In order to simplify the use of position control function in user applications, STEP7 -- Micro/WIN provides a position control guide can help users in a very short period of time to complete the PWM、PTO or position control module configuration. The guide can generate position command, the user can use these instructions in the application provide dynamic control for speed and position.

With the help of position control guide configuration PTO output, require the user offer some basic information, the output of the guide configuration regarding requires the user to provide some basic information, item by item is introduced as follows:

a、Maximum（MAX_SPEED）and start/stop speed（SS_SPEED）

Figure 2-3 is the sketch map of these two concepts. MAX_SPEED is the allowance max. value of operation speed, it should be in the range of motor moment capacity. The moment which required by the drive load is decided by friction、inertia and time of accelerate/decelerate.

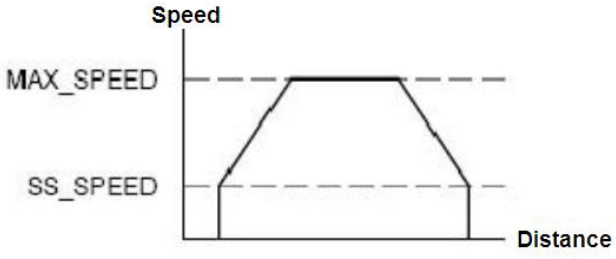


Figure 2-3

The value of SS_SPEED should be satisfied with the capacity that drive the load during motor with low speed, if SS_SPEED value is too low, at the start and stop time, the motor and load maybe swing or vibration. If SS_SPEED value too high, motor lose the pulse when start, and when the load try to stop will make motor over speed. Usually, SS_SPEED value is 5%-15% of MAX_SPEED.

b、Speed up time and deceleration time

Speed up time ACCEL_TIME: the required time that motor from SS_SPEED speed up to MAX_SPEED speed. Deceleration time DECEL_TIME: the required time that motor from MAX_SPEED speed down to the SS_SPEED speed.

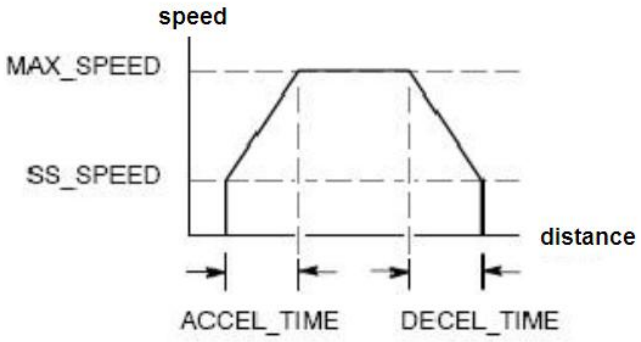


Figure 2-4

The default setting of acceleration time and deceleration time are both 1000 milliseconds. Normally, motor can work within 1000 milliseconds times. Reference Figure 2-4, these 2 value setting should be take milliseconds as the unit. The acceleration speed or underspeed of motor normally need the test to confirm. At the beginning, it should input one bigger value. Drop off the time value gradually until motor start lose speed, so optimize these setting.

c、Moveable envelope

An envelope is a pre-defined mobile description, which includes one or more speed, affects the move from start to finish. An envelope is composed of multiple stage, each section contains one acceleration/deceleration process that up to the target speed and a pulse with a string of fixed quantity which average running based on a target speed.

Position control guide provides mobile envelope defined interface, each movable envelope required for the application program are all can be defined here. PTO support maximum 100 envelope.

Define an envelope, including the following points: (1) choose operating modes; (2) Define index for each step envelope (3) Defines a symbolic name for the envelope.

(1) Choose operation mode of envelope: PTO support two modes, continuous rotation of

relative position and single speed, Figure 2-5 for reference.

Relative position mode refers to the final position of movement calculate pulse quantity from the starting point side. Single speed continuous rotation no need to offer final point position, PTO always continuous output pulse, until there is other instruction send out.

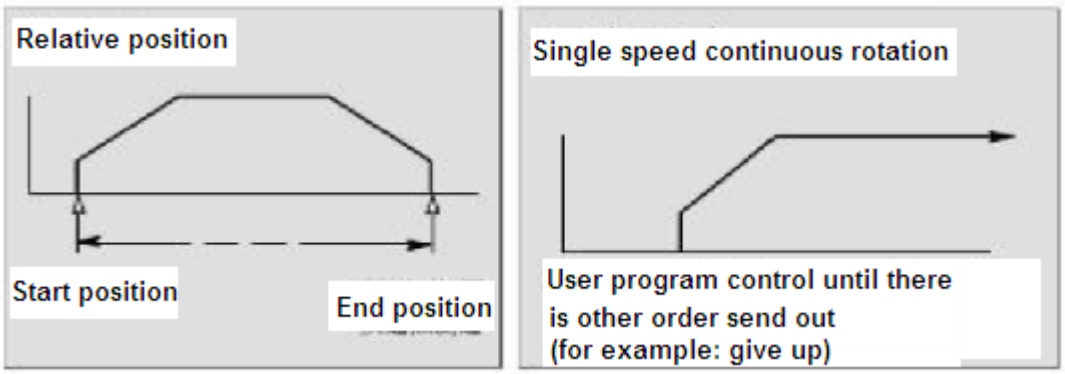


Figure 2-5

⑵ Step in envelope

A step is a fixed distance in work-piece movement, including the distance that in the acceleration and deceleration time. PTO each envelope maximum allows 29 steps. Each step includes target speed and finish position or pulse quantity etc.

Figure 2-6 show one step、two step、three step and four step envelope.

Note that one step envelope only have one normal speed segment, two step envelope have two normal speed segment……The number of steps same as the number of normal speed in the envelope.

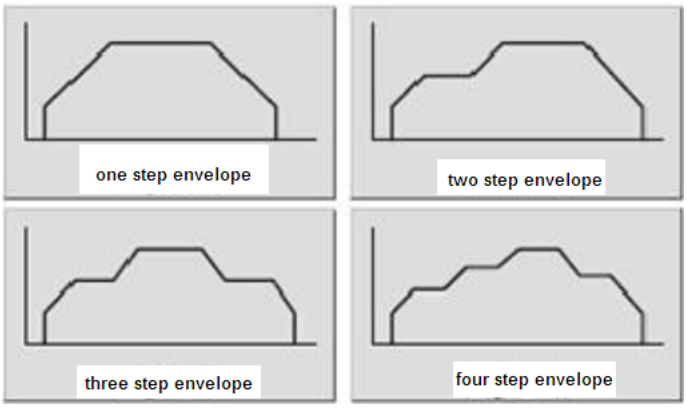


Figure 2-6 step no. sign of envelope

2.3 Electrical introduction

2.3.1 Proximity switch



Proximity switch is one kind of position switch that no need directly contact with the moving parts, when metal detection part approaching to switch induction surface to movement distance, don't need a mechanical contact and also do not need make any pressure, then can make switch action, and thus drive DC electrical or offer control instruction to computer (PLC) device. The working principle diagram as shown below

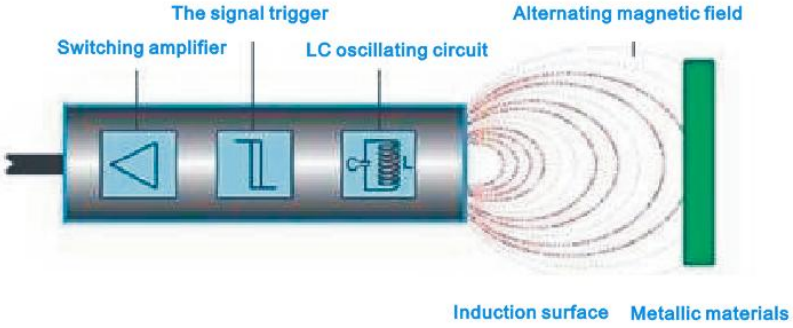


Figure 2-7 Proximity switch working principle diagram

The proximity switch used in this XK-JS1-1 is NPN type three wires system type: Red (brown) wire connects plus end of power; blue wire connect power 0V end; yellow (black) wire is signal, should connect load. Another end of load connection as following: for NPN proximity switch, should connect power plus end; for PNP proximity switch, should connect power 0V end. In this device, use the proximity switch as origin signal of X axes and Y axes.

2.3.2 Micro-switch



Micro-switch with tiny contact interval and fast mechanism, with the provisions stroke and force to do the dynamic action, cover with shell, outside with one drive rod, because its contactor spacing is small, so called micro-switch, through driving element, the outer mechanical force act the force on the action reed, when action reed displace to critical point and produce split-second response, it make the movable contact and constant contact at the end of action reed quick connect or break. This XK-JS1-1 device total use four micro-switch, they are X axes left limit、X axes right limit、Y axes frontward limit and Y axes backward limit.

2.4 IO Distribution List

Table 2-1 IO distribution list

Type	IO	Meaning
Input	I0.0	Z-axis origin sensor
	I0.1	X-axis befor limit
	I0.2	X-axis after limit
	I0.3	Y-axis up limit
	I0.4	Y-axis down limit
	I0.5	Auto button
	I0.6	Reset button
	I0.7	Grasp/Start button
	I1.0	Left handed/Stop button
	I1.1	Right hand button
	I1.2	Forward button
	I1.3	Back button
	I1.4	Up button
	I1.5	Down button
Output	Q0.0	Z-axis pulse output
	Q0.1	XY-axis pulse output
	Q0.2	Z-axis motor dir
	Q0.3	X-axis motor dir
	Q0.4	Y-axis motor dir
	Q0.5	X-axis pulse output
	Q0.6	Y-axis pulse output
	Q0.7	Claw grasping

2.5 PLC wiring principle diagram

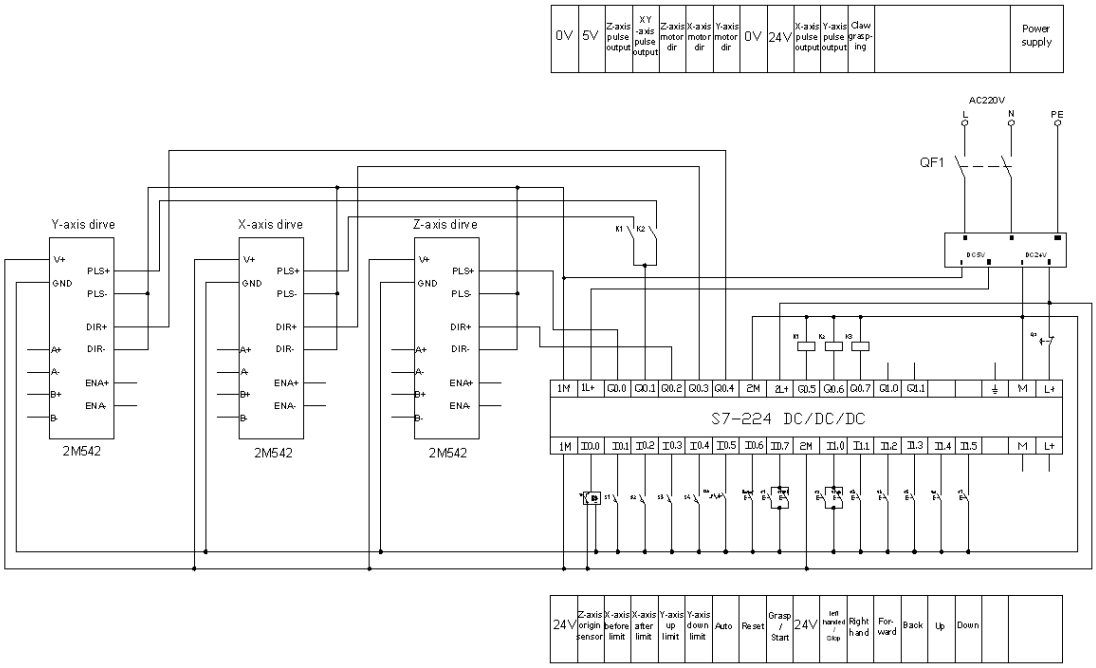


Figure 2-6 PLC wiring principle diagram

2.6 System operation demo

This system has two operation modes: manual operation and automatic operation.

Manual operation can control the X axle、Y axle、Z axle and inching operation control of claw. Automatically running can make the manipulator carrying things automatically. Users can do PLC programming, according to user requirement to realize different motion control, user programmable IO table shown in Figure 2-1, Wiring diagram as shown in Figure 2-6

First to power supply for device, plug into the AC220V socket, open the leakage protector switch, power supply for equipment, unscrew the emergency stop button on the button box, to power for PLC.

Manual running operation: Turn the QA two position selector switch to "Manual" (that is mean: the left side), press "S1" button, now the claw is grasping, loosen the button, the claw is loosened; press "S2" button, now the manipulator turning left, loosen the button, the manipulator stop movement; Press "S3" button, now the manipulator turn right, loosen this button, manipulator stop movement; Press "S4" button, the manipulator go forward, loosen this button, manipulator stop movement; Press "S5" button, the manipulator go backward, loosen this button, manipulator stop movement; Press "S6" button, the manipulator goes upward, loosen this button, the manipulator stop movement; Press "S7" button, the manipulator goes downward, loosen this button, the manipulator stop movement.

Automatic running operation: before the automatic running, it should be do the manual reset operation for Z axis, under the manual operation state, use buttons of "S2" and "S3" to make the manipulator movement to the origin position (i.e., Z axis origin sensor with signal position),

Turn QA two position selector switch to "Auto" (that is, the right side), press the "Reset" button, the X axes and Y axes of system automatic reset, reset is completed, press the "Start" button, the system automatically running, namely carry the goods from the right front position of the training table to the left rear position, then automatically return, begin the next round of carrying.

In the process of carrying, if press the "Stop" button, after the system completes this time of carrying process, Stop moving after back to the origin. If require to restart automatically, press the "Start" button, means entre into the automatically running once again.

Attachment A Terminal strip wiring diagram

